Universal Windows Platform with Mobile and Desktop extensions.

With such future devices it will be important for applications to adapt at runtime from a mobile form factor, to a desktop form factor, to even a television form factor. For this Microsoft has provided new adaptive UI elements that can work across form factors. Developers can also embrace the principles of responsive design for desktop applications – already a popular approach for web apps.

Universal Windows Platform Bridges

It's quite a mouthful, but these bridges – as we will refer to them as for brevity – are toolkits designed to make it easier for developers to bring their existing Android, iOS, Web, and even old Windows apps and development experience to Windows 10 and to sell such apps on the Windows store.

These bridging technologies are: Project Astoria for Android, Project Islandwood for iOS, Project Westminster for web applications, and Project Centennial for 'Classic Windows apps'.

Microsoft has yet to fully explain how these work, however they claim at least for Android that existing applications will need few code changes. For cases where your application uses Google services Microsoft has an interoperability layer that can map those functions to Microsoft's technologies. For instance, it has API mappings for Ads, Analytics, In-app purchases and Notifications. With minimal code changes it can make your code automatically use Bing Maps on Windows while using Google Maps on Android.

These bridges also give developers access to Windows features such as Live tiles while still being able to target competing platforms. In fact, you can even target Windows Phone using your IDE on an OS X system.

Essentially this means Android developers can continue to code in Java, iOS developers can continue to code in Objective-C (they plan to support Swift as well), web developers can continue to code in web technologies and Windows developers used to older technologies can still continue to use them. And they can still have their application running on Windows platforms and sold on the Windows store.

It's already possible to start using their compatibility layer for web applications, which allows you to have websites present as native apps in native windows. Developers can sign up for access to the other compatibility layers while they are still in development.

A Consolidated and Enriched Store

Gone are the platform-specific stores, with Windows 10 there is a single store for all platforms running Windows, from the Raspberry Pi to the Xbox. The same store now also offers other kinds of media such as music, and videos. For developers this also means there is now a single place to submit apps to, and the possibility of submitting a single package that can run on all platforms. The Windows store is also no longer restricted in the way it was in Windows 8. It can now support traditional Win32 applications through the compatibility layer mentioned above.

Interactive page

To keep users secure, apps installed from the Windows store, are sandboxed from each other and can only talk to each other though APIs designed for the purpose. One app cannot interact with another apps' settings or private data, and in fact gets its own private virtual registry. This makes it harder for malware to do anything too damaging while also making it possible to quickly and consistently uninstall apps in a way that gets rid of all their data.

The features of this unified app store have also been enhanced with the ability to offer apps on a subscription model, and to have per-market pricing. The platform is now smart enough to share libraries between different applications if they use the same version, thus reducing disk space and download sizes.

There are also interesting implications of the new constantly updating model of Windows. New updates to Windows 10 can bring new platform features and APIs, however any updates to these APIs are versioned. So when developing an application, you can specify the range of versions of Windows that your application supports and the device families it supports (desktop, mobile, holographic, IoT etc.). If your app runs on the oldest version of the API and uses only UWP it should run on all platforms and devices; although in practicality such apps will be few.

This is quite similar to Android's way of doing things where you can select the oldest version of Android you want to target, and that will lock you to the subset of APIs that that device supports. Another similarity with Android is the support for universal binaries. Just like it's possible to have a single Android apk that runs on any Android device be it a mobile, tablet or TV, it is now possible to have single Windows binary that that can run on any platform that supports Windows, even if that platform didn't even exist when you compiled it!

Another treat for those developing for the Windows store is the ability to hide apps pushed to the store so they can only be accessed if you know the URL. Windows Store will also soon be available as a website that people can browse and buy apps from.

Other Bits and Bobs

.NET Native: For performance conscious .NET application developers, there is some good news,

*pointers

>>Interesting developer news

***Visual Studio 2015 Keynote**

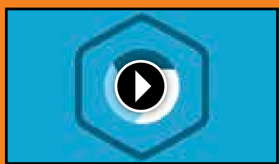
>>Watch an overview on how to get started with using Visual Studio 2015, .NET 4.6 and Visual Studio Online.

http://dgit.in/VisStudio2015

***Intel's IoT Developer Kit**

>>This video gives you an introduction to Intel's IoT Developer Kit for Intel Galileo and Intel Edison boards.

http://dgit.in/IoTDevKit



***Working on Chrome Developer Tools**

>>Watch this short video series on using the Chrome Developer Tools by phpacademy.

http://dgit.in/ChrmDevTools

***Google's Deep Dream Project**

>>This video tells you what Google's Deep Dream is all about and how it works.

http://dgit.in/DeepDream